INCH-POUND

MIL-DTL-17587/4(SH)
28 June 2007

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, ACB, LOW VOLTAGE, ELECTRIC POWER, AIR, REMOVABLE CONSTRUCTION ACB-5KV-2000RC

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17587.

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REQUIREMENTS:

TABLE I. Ratings/features of circuit breakers.

Circuit breaker type	**ACB-5KV-2000RC**
Rated continuous current (amperes)	2000
Rated maximum voltage (AC volts) /1	4160 /3
Frequency (hertz)	60
Maximum interrupting rating current (symmetrical amperes)	56,000
Rated short-time current /2 (amperes) (see 6.3.18)	80,000
Rated short-time current duration (seconds)	0.16
Operating mechanism - manual	STD
Operating mechanism - electric	STD
Maximum weight (lbs) - circuit breaker (removable component)	540
Maximum weight (lbs) - cradle (stationary component)	190
Number of poles	3
Number of overcurrent coils or solid-state sensors	3
Attachments	
Electric closing mechanism	
Auxiliary switches (number of contacts)	10
Undervoltage lockout device	STD
Mechanical position indicator	STD
Secondary disconnecting devices	STD
Cell switch	STD
Indicator light	STD
Hold-closed mechanism	STD
Undervoltage release device	STD
Shunt trip device	STD
Operations counter	STD
Trip on power up (TOPU) device	N/A
Rack-in/rack-out position indicator	STD

NOTES:

/1 Voltage ratings apply to main current-carrying parts. Operating attachments shall be of voltages specified in 6.2.

/2 Ampere values apply to circuit breakers with overcurrent coils or sensors of maximum current rating within the circuit breaker type.

/3 When specified (see 6.2), circuit breaker devices (such as shunt trip and overcurrent trip) shall meet requirements over a range from 15 to 60 Hz.

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)						Maximum instantaneous trip /2 (amperes)
ACB-5KV-2000RC (ODS breaker box)	25	37.5	50	62.5	50	75	100	112.5	125	----	500
	50	75	100	125	100	116.5	150	200	225	250	2000
	100	150	200	250	200	300	400	450	500	----	2000
	200	300	400	500	400	600	800	900	1000	----	4000
	300	450	600	750	600	900	1200	1350	1500	----	6000
	400	600	800	1000	800	1150	1200	1400	1600	1800	8000
	500	750	1000	1250	1000	1500	2000	2250	2500	----	10000
	600	900	1200	1500	1200	1800	2400	2700	3000	----	12000
	700	1050	1400	1750	1400	2100	2800	3150	3500	----	14000
	800	1200	1600	2000	1600	2400	3200	3600	4000	----	16000
	1000	1500	2000	2500	2000	3000	4000	4500	5000	----	20000
	1200	1800	2400	3000	2400	2800	3600	4800	5400	6000	24000
	1400	2100	2800	3500	2800	4200	5600	6300	7000	----	28000
	1600	2400	3200	4000	3200	3500	4800	6400	7200	8000	32000
	1800	2700	3600	4500	3600	5400	7200	8100	9000	----	36000
	2000	3000	4000	5000	4000	6000	800	9000	10000	----	40000

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TABLE II. Available continuous current ratings and pickup current settings for overcurrent coils and sensors - Continued.

Type	Coil or sensor rating /1 (amperes)	Long-time pickup setting (amperes)			Short-time pickup setting (amperes)						Maximum instantaneous trip /2 (amperes)
ACB-5KV-2000RC (EOC breaker box)	100	150	160	165	218.8	233.3	265	287.5	300	350	2000
		169	180	200	400	450	500	---	---	---	
		210	250	---	---	---	---	---	---	---	
	200	300	320	330	437.5	466.6	530	575	600	700	4000
		338	360	400	800	900	1000	---	---	---	
		420	500	---	---	---	---	---	---	---	
	300	450	480	495	656.4	699.9	795	862.5	900	1050	6000
		507	540	600	1200	1350	1500	---	---	---	
		630	750	---	---	---	---	---	---	---	
	400	600	640	660	875.2	699.9	795	862.5	900	1050	6000
		676	720	800	1200	1350	1500	---	---	---	
		840	1000	---	---	---	---	---	---	---	
	1200	1800	1920	1980	2625.6	2799.6	3180	3450	3600	4200	24000
		2028	2160	2400	4800	5400	6000	---	---	---	
		2520	3000	---	---	---	---	---	---	---	
	1600	2400	2560	2640	3500.8	3732.8	4240	4600	4800	5600	32000
		2704	2880	3200	6400	7200	8000	---	---	---	
		3360	4000	---	---	---	---	---	---	---	

NOTES:

/1 Continuous current rating.

/2 The instantaneous trip device shall be set at the place of manufacture from five times the coil or sensor rating to the maximum pickup setting specified for mechanical coils and 20 times sensor rating for solid-state sensors with a default at maximum listed rating.

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TABLE III. Electrical endurance, temperature rise, creepage and clearance, dielectric strength, and insulation resistance test requirements.

Electrical endurance test:

Type	Total number of close-open cycles	Amperes on circuit breaker close	Amperes on circuit breaker open	Maximum a.c. power factor (lagging) X/R
ACB-5KV-2000RC	4,000 (Arc chute replacement shall be permitted after 1000 close-open cycles)	8,000	2,000	Circuit breaker close: 0.25 Circuit breaker open: 0.80

Temperature rise test:

Type	Continuous current rating	Cable quantity per terminal /1	Cable size	Bus bar quantity per phase	Bus bar size (inches) /2
ACB-5KV-2000RC	2,000	5	400 MCM	2	8.0 x ¼

Creepage and Clearance:

Type	Creepage	Clearance
ACB-5KV-2000RC	4 inches on top surfaces 3 inches on bottom and side surfaces	3 inches on all live parts to ground 2.6875 inches across main contacts

Dielectric Strength and Insulation Resistance:

Type	Dielectric Strength	Insulation Resistance
ACB-5KV-2000RC	Circuit breaker insulation in as-new condition shall be able to withstand without breakdown a 60 Hz voltage of 19,000 volts for a period of 1 minute when applied in accordance with 4.5.11.3.1.1 and Tables XIV and XV of MIL-DTL-17587. As circuit breaker is cycled and tested for qualification, dielectric strength requirements at 60 percent of value above may apply. See 4.5.11.2 of MIL-DTL-17587 for these test requirements.	Testing shall be conducted in accordance with 3.4.10.4 and section 4.5.12 of MIL-DTL-17587 except that insulation resistance shall be measured with a 4160-volt insulation resistance testing device.

NOTES:

/1 Information regarding cable is contained in MIL-DTL-915. If 3-phase cable is used for this test, the cable jacket shall be stripped for a minimum of 2 feet.

/2 Information regarding bus bars is contained in MIL-DTL-16036.

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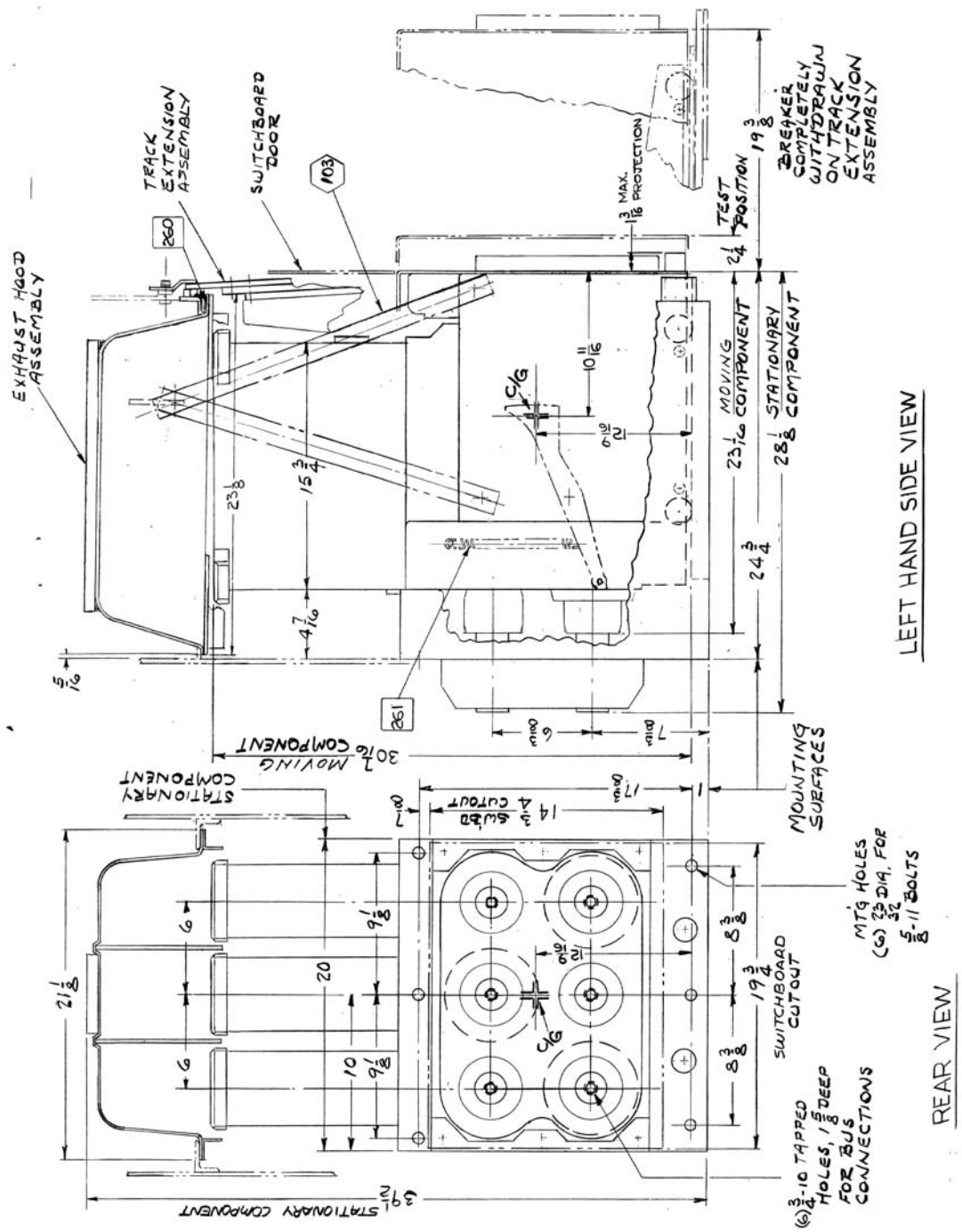


FIGURE 1. Outline and mounting dimensions for circuit breaker type ACB-5KV-2000RC.

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SPECIFIED CALIBRATION AT 60 Hz

LONG TIME DELAY

12.8 – 18.5 SECONDS AT 225% COIL RATING
(L.T.D. PICK-UP AT 150% COIL RATING)

8.4 - 12.1 SECONDS AT 225% COIL RATING FOR LTS. (WHEN ORDERED)

SHORT TIME DELAY

CURRENT % of P.U.	TIME BAND (Seconds)			
	MINIMUM (1)	INTERMED. (2)	MAXIMUM (3)	SP. MAX. (4)
250	.064 - .12	.16 - .27	.32 - .42	.47 - .64

(BASED ON S.T.D. PICK-UP OF 400% OF COIL RATING)

NOTE: TIME BAND DESIGNATES RESETTABLE DELAY & TOTAL CLEARING TIME CORRESPONDING TO EACH TIME BAND.

4. SPECIAL MAXIMUM BAND

3. MAXIMUM BAND

2. INTERMEDIATE BAND

1. MINIMUM BAND

INSTANTANEOUS TRIP

PICK-UP SHOWN SET AT 2000 % OF COIL RATING

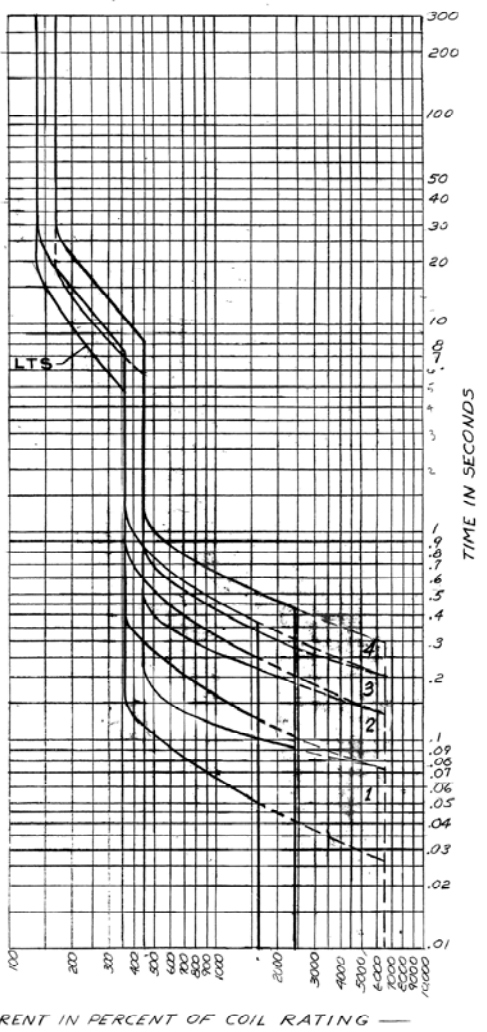


FIGURE 2. Time-current characteristics curve for circuit breaker type ACB-5KV-2000RC (ODS and EOC versions) at 60 Hz.

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15 HZ RESPONSE WHEN CALIBRATED AT 60 HZ

LONG TIME DELAY

11.2 - 14 SECONDS AT 225 % COIL RATING
(L.T.D. PICK-UP AT 106 % COIL RATING)
7.3 - 10.4 SECONDS AT 225 % COIL RATING FOR LTS (WHEN ORDERED)

SHORT TIME DELAY

CURRENT % OF P.U.	TIME BAND (SECONDS)			
	MINIMUM (1)	INTERMED. (2)	MAXIMUM (3)	SP. MAX. (4)
250	.09 - .17	.22 - .32	.45 - .55	.60 - .82

(BASED ON S.T.D. PICK-UP OF 450 % OF COIL RATING)

NOTE: TIME BAND DESIGNATES RESETTABLE DELAY & TOTAL CLEARING TIME CORRESPONDING TO EACH TIME BAND.

4. SPECIAL MAXIMUM BAND.

3. MAXIMUM BAND

2. INTERMEDIATE BAND.

1. MINIMUM BAND

INSTANTANEOUS TRIP

PICK-UP AT 3000 % OF COIL RATING

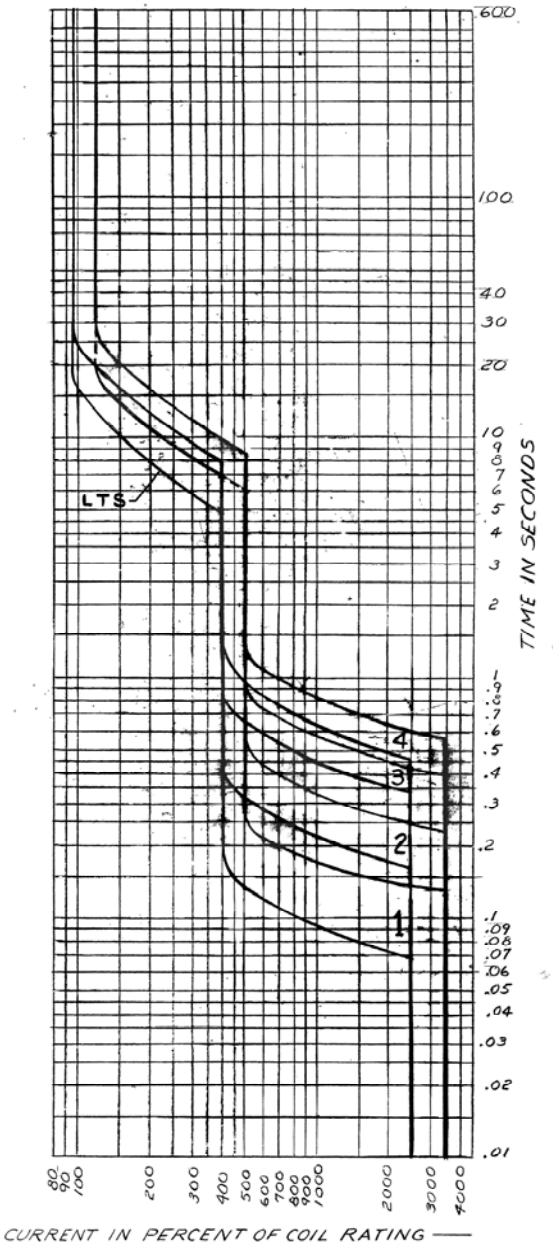


FIGURE 3. Time-current characteristics curve for circuit breaker type ACB-5KV-2000RC (ODS version) at 15 Hz.

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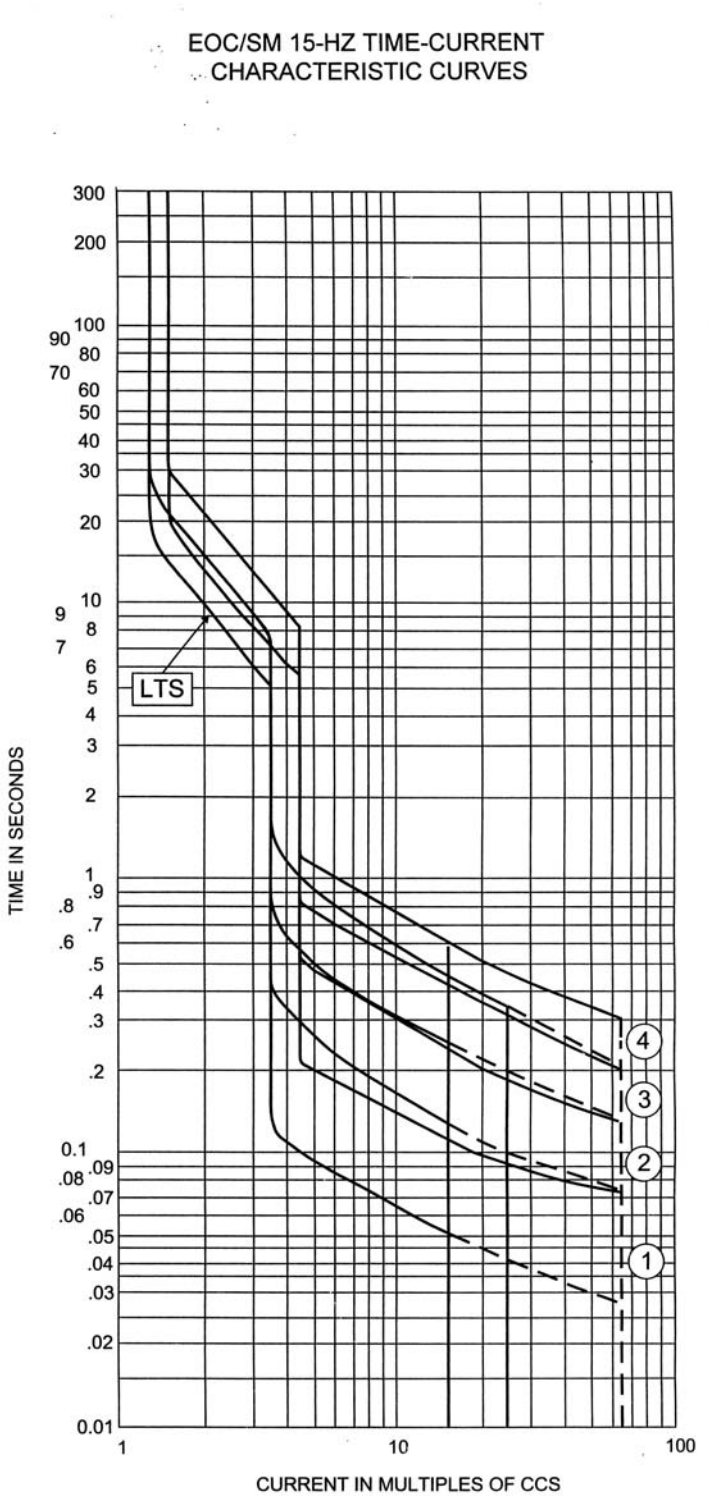


FIGURE 4. Time-current characteristics curve for circuit breaker type ACB-5KV-2000RC (EOC version) at 15 Hz.

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Preparing Activity:
Navy – SH
(Project 5925-N184-004)

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